

One Coalition, Two Electorates

A short, code-free tour of what the data showed about Reform UK's voters

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The question

There's a comfortable shorthand in political commentary: "the Reform voter." I wanted to know whether that single label hides more than one kind of person.

Put precisely, the question the project sets out to answer is this: **is Reform UK's electoral coalition attitudinally unified, or does it contain a substantial cross-pressured subgroup whose economic and social attitudes sit in tension with the party's own positions?** That phrasing matters. It asks about structure inside the electorate, not about who will win anything, and it can be answered with attitudes alone—without ever showing the model who voted for whom.

The starting idea is **cross-pressure**, a classic in electoral sociology. Lazarsfeld, Berelson and Gaudet (1944) described voters pulled in different directions by conflicting social influences, who resolve the tension by deciding late, holding their party loosely, or voting in ways that sit oddly with their stated views. Hillygus and Shields (2008) recast it around issues: a voter whose position on something salient diverges from their party's platform is more persuadable and more likely to carry internally inconsistent bundles of attitudes. That second version is the one this project tests.

The argument I'm walking into

This is a live disagreement, and it's worth knowing the shape of it before looking at any results.

Ford and Goodwin (2014) brought cross-pressure to the British radical right directly, describing UKIP's base—Reform's predecessor—as “left behind”: older, working-class, economically insecure voters who pair socially authoritarian attitudes with a real attachment to state support. It is the thesis most people are repeating, usually without knowing they're repeating it.

Evans and Mellon (2016) contest it. Drawing on long-run panel data, they argue the radical-right vote is better explained by *composition*—which voters defected, and in what order—than by a stable economically-left core. Their objection is partly about evidence: much of the left-behind case rests on aggregate patterns, and aggregate patterns can mislead about individuals.

Norris and Inglehart (2019) push in a third direction entirely. Their *cultural backlash* thesis treats radical-right support as a reaction against progressive value change rather than material grievance—values, not economics.

Sobolewska and Ford (2020), in *Brexitland*, then reframe the original thesis from the inside, moving its centre of gravity from economic left-behindness toward an identity divide driven by the expansion of higher education. Education, on this account, is not a proxy for income; it is where the values themselves are formed.

And the most recent descriptive baseline, NatCen's *British Social Attitudes 43: Who Supports Reform?* (2026), profiles Reform's voters using the same survey series—but describes them without clustering them, and doesn't take up the cross-pressure question at all.

So there is a well-developed argument about what this electorate *is*, conducted largely with descriptive statistics and panel data. What nobody had done was ask an unsupervised model whether the tension shows up inside Reform's own voters, without telling it what to look for. That's the gap this project works in—and because the four positions above make different predictions, the data can actually speak to them rather than just illustrating one.

Where the data made me change course

Three honest moments shaped the result, and I've left them visible rather than tidied away.

A plan the data wouldn't support. I'd intended to study LGB Reform voters and their own equality attitudes. But the survey's split-ballot design meant that of 336 Reform voters, only fifteen identified as non-heterosexual, and just two of those were asked the relevant questions. Two is not a finding. So I reframed the question from *identity-versus-vote* to *attitude-versus-attitude*—the stronger question anyway.

A judgement call over a metric. After using PCA to find the main attitudinal axes, I clustered the voters. The usual metric (the silhouette score) was highest at two clusters—but two clusters lumped almost all Reform voters together and hid exactly what I was looking for. I chose three, trading a little statistical neatness (0.392 to 0.358) for a result that answered the question. As it happens, the two-cluster optimum *is* the two-electorates story; the third cluster is just a small left-libertarian fringe.

A method that disagreed with itself. A first autoencoder seemed to confirm the clusters neatly—until I noticed its result leaned on a geometric quirk. A second one, built to remove the quirk, disagreed. Instead of quietly keeping the flattering number, I followed the disagreement, and it led to the most useful part of the analysis: pinning down exactly how much to trust the split.

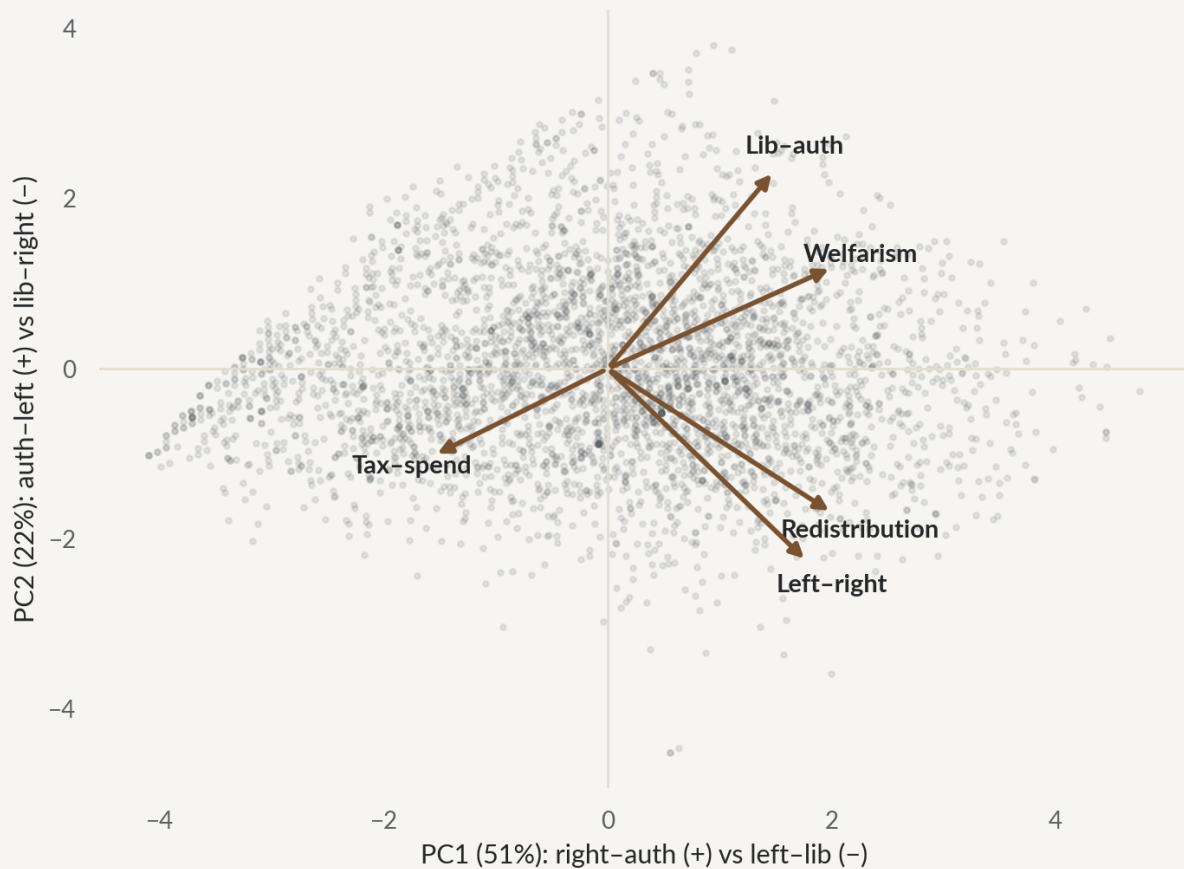
The two axes the electorate divides on

Before the clusters, the axes. Principal component analysis on five attitude scales returns two components holding about three-quarters of the variance. The first is the familiar left-right dimension that fuses social and economic attitudes. The second is the interesting one: it separates people who are *authoritarian and*

economically left at the same time from their libertarian-right opposites. That second axis is what the whole project is built around, and it exists in the data before any clustering is done.

The two axes read as left–right and authoritarian–libertarian

PCA biplot: loadings of the five attitudinal items on the first two components



Source: British Social Attitudes 2024 (UK Data Service). Analysis: B. Quinlan.

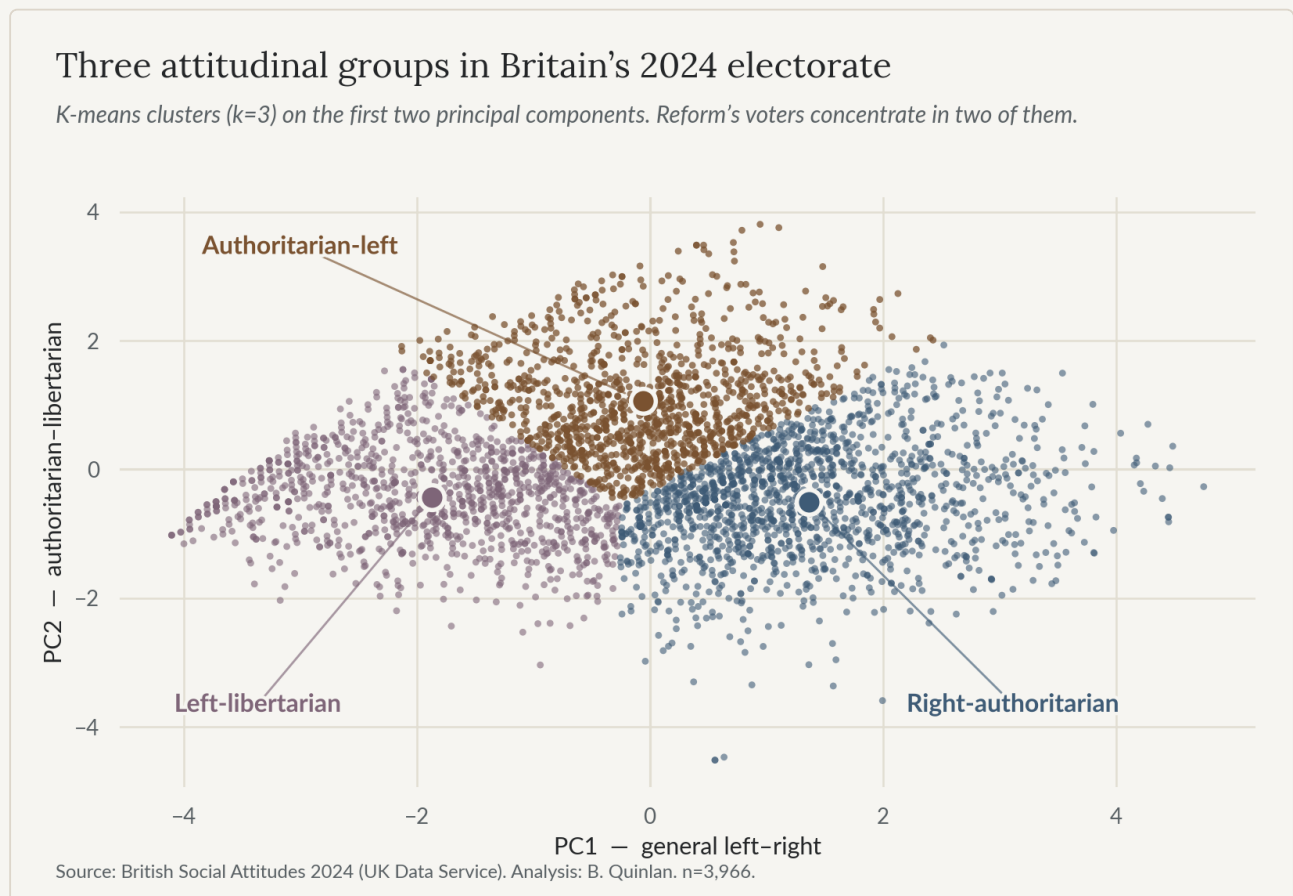
The two attitudinal axes: PC1 (left–right fusion) and PC2 (authoritarian-left vs libertarian-right)

The finding: two electorates, not one

Clustered on political attitudes alone, with no sight of vote choice, Reform's voters split almost in half:

- 55% sit in a conventional **right-authoritarian** group.
- 41% sit in an **authoritarian-left** group—socially hardline, but economically left.
- 4% fall in a small left-libertarian cluster that is otherwise almost Reform-free.

Those are shares of the 329 Reform voters in the 2024 British Social Attitudes Survey, located inside a sample of 3,966 respondents.



The three-cluster solution: ~55% right-authoritarian, ~41% authoritarian-left

That second group is the cross-pressured one. Their economic instincts sit in real tension with the party they voted for—Reform campaigned in 2024 on tax cuts it costed itself at nearly £90bn a year, a figure the IFS reported and judged an underestimate (Emmerson, Joyce and Miller, 2024).

What kind of group is it?

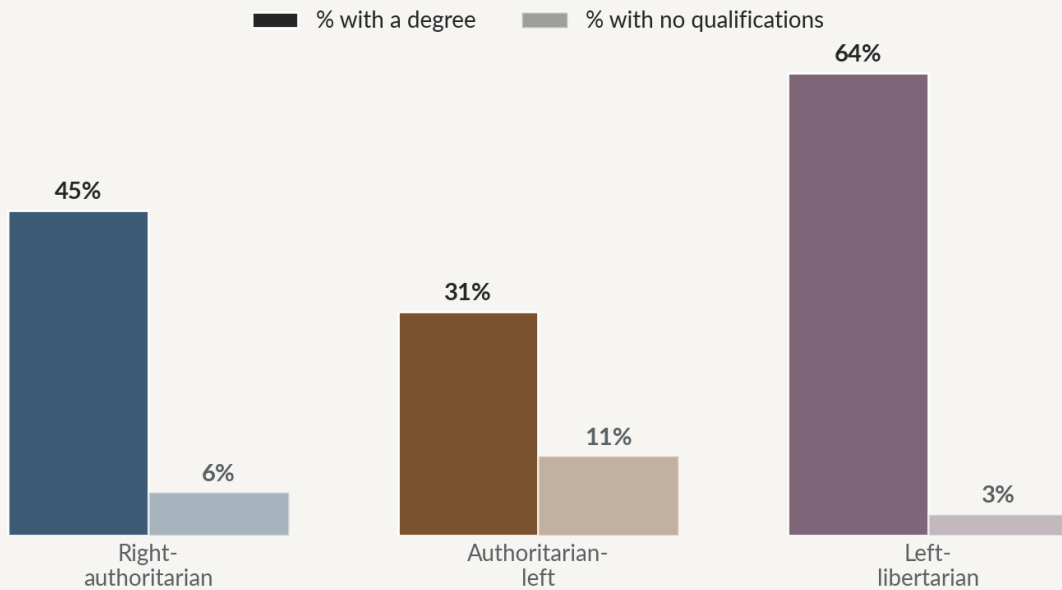
“Economically left” undersells what’s going on, and getting this right took some care.

This group is pro-redistribution *and* about as sceptical of welfare claimants as the conventional right-authoritarian group *and* nearly as anti-immigration (around four in five, on the relevant split-ballot items, rate immigration bad for the economy). It decouples two things most voters keep welded together: taking from the better-off, which it supports, and sympathy for claimants, which it doesn’t. Redistribution aimed at an implied in-group, held alongside hostility to an out-group’s claim on the same welfare state, is what political scientists call **welfare chauvinism** (Andersen and Bjørklund, 1990)—a more precise description, and a less paradoxical one, than “a left-wing electorate inside a right-wing party.”

What makes the “left-behind” label concrete is **income and education**. This group’s Reform voters sit a full household-income band below the conventional-right group’s, and they are much the least qualified of the three—31% hold a degree, against 45% and 64%. That education gap holds even after adjusting for social class and household income, which is what *Brexitland* predicts and what a purely economic account does not.

The authoritarian-left cluster is the least formally educated

Highest qualification held, as a share of each attitudinal cluster (solid = degree, faded = none)



Source: British Social Attitudes 2024 (UK Data Service). Analysis: B. Quinlan.

Formal education tracks the divide: the authoritarian-left cluster is the least qualified

I had expected benefit receipt to do that work, and checked it properly rather than assuming. It doesn't. The survey's benefit question counts the state pension, winter fuel payments and child benefit, so for an old-skewing electorate it partly measures age: above 65 the two Reform groups are indistinguishable. Benefit receipt ended up the weakest claim in the project, and saying so is more useful than leaning on it.

Is it real—and is it old?

How much to trust it. The autoencoder that disagreed taught me where the uncertainty really lives. The split barely moves if I drop the weakest variable, cluster on the full data instead of a 2-D summary, or resample the respondents—a Reform voter placed in the left-behind group stays there across resamples 98% of

the time. Hierarchical clustering agrees strongly with the main method, and a decision tree, coming at the problem backwards, finds the same dominant attitude and—tellingly— struggles to pick out the cross-pressured group, exactly as a genuinely conflicted group should. The one thing that *can* redraw the split is a radically different way of compressing the data, and even then only at the soft border between the two Reform groups, never in whether the left-behind group exists.

How old it is. Projecting 2014 UKIP identifiers into the same model, they fall into the “left-behind” cluster *even more heavily* than 2024 Reform voters—59% against 41%, a gap well beyond chance. The pattern predates Reform by a decade.

But the raw share is the wrong quantity for the sharper question, because the whole population’s authoritarian-left share also shrank over the decade.

Measured against each year’s own population, the left-behind group is just as over-represented in 2024 as in 2014, while the conventional-right group flips from *under*-represented to *over*-represented. Reform’s coalition didn’t move away from its left-behind core; it added a conventional-right layer on top of one that was already there. That is the distinction *Brexitland* draws between an electorate changing its mind and a coalition being recomposed—and the pattern here fits the second.

What I’m claiming—and what I’m not

The claims are deliberately **descriptive**, not predictive. These are structural patterns in attitudes, not turnout forecasts, and nothing here says how any particular constituency will vote.

Four limits are worth stating plainly, because a careful reader will reach for them.

The subgroups are small. 329 Reform voters, and the split-ballot design confines the immigration and LGB items to roughly a quarter of the sample. Those figures profile the clusters; they don't define them, and they're reported with their sample sizes throughout.

The clustering is unweighted. Fitting survey weights inside an unsupervised pipeline is non-standard, so the percentages describe the analytic sample rather than the population. As a check, reweighting the final split to the survey's published weights moves it only to 52/44/4 against 55/41/4—so the two-electorates finding isn't an artefact of that.

Benefit reliance is the weakest claim in the project, for the reasons above, and the left-behind reading rests on income and education instead.

A decade apart is not a panel. Two independent cross-sections can show that the profile is durable and that the coalition's composition shifted. They cannot show that any individual voter changed their mind. Separating conversion from composition at the individual level would need panel data such as the British Election Study—which is precisely the evidence Evans and Mellon bring to the original debate, and precisely what a repeated cross-section cannot settle.

What the analysis does establish is structure and its persistence—across linear and non-linear methods, supervised and unsupervised, and across ten years of data. A single “Reform voter” label hides two quite different electorates, and the cross-pressured half is real, robust, and at least a decade old.

Data: British Social Attitudes Survey 2024 (UK Data Service SN 9478) and 2014 (SN 7809), NatCen Social Research; English Indices of Deprivation 2019 (MHCLG); 2025 Local Elections Handbook (House of Commons Library). The raw survey microdata is licence-restricted and not included in this repository—see [data/README.md](#).

Works cited

Andersen, J.G. and Bjørklund, T. (1990) 'Structural Changes and New Cleavages: the Progress Parties in Denmark and Norway', *Acta Sociologica*, 33(3), pp. 195–217.

Emmerson, C., Joyce, R. and Miller, H. (2024) *Reform UK manifesto: a reaction*. London: Institute for Fiscal Studies. Available at: <https://ifs.org.uk/articles/reform-uk-manifesto-reaction> (Accessed: 6 August 2026).

Evans, G. and Mellon, J. (2016) 'Working Class Votes and Conservative Losses: Solving the UKIP Puzzle', *Parliamentary Affairs*, 69(2), pp. 464–479.

Ford, R. and Goodwin, M.J. (2014) *Revolt on the Right: Explaining Support for the Radical Right in Britain*. Abingdon: Routledge.

Hillygus, D.S. and Shields, T.G. (2008) *The Persuadable Voter: Wedge Issues in Presidential Campaigns*. Princeton: Princeton University Press.

Lazarsfeld, P.F., Berelson, B. and Gaudet, H. (1944) *The People's Choice: How the Voter Makes Up His Mind in a Presidential Campaign*. New York: Columbia University Press.

NatCen Social Research (2026) *British Social Attitudes 43: Who Supports Reform?* London: NatCen Social Research.

Norris, P. and Inglehart, R. (2019) *Cultural Backlash: Trump, Brexit, and Authoritarian Populism*. Cambridge: Cambridge University Press.

Sobolewska, M. and Ford, R. (2020) *Brexitland: Identity, Diversity and the Reshaping of British Politics*. Cambridge: Cambridge University Press.

The full reference list is in [report.html](#) .